1. EXECUTIVE OVERVIEW & PROBLEM STATEMENT

Business Context: An online marketplace processes transactions in MySQL (orders, customers, products), clickstream events in web logs, and support tickets in text files. Marketing lacks a unified customer view, leading to unrecovered abandoned carts and delayed personalization.

Core Problem Statement: Build a Customer 360 platform unifying transactions, clickstream, and support tickets to detect cart abandonment in real-time (<60s alert), store unified profiles in HBase, and support natural language RAG queries.

2. CORE BUSINESS KPIS & TECHNICAL SLAS

Metric Type	Key Performance Indicator (KPI)	Target Benchmark	Business Impact
Business	Abandoned Cart Recovery Rate	**>= 18% recovered**	Directly boosts top-line revenue from idle carts
Business	Customer Lifetime Value (LTV)	**+12% increase**	Enhances customer retention and cross-sell
Operational	HBase Customer 360 Lookup	**< 100 ms latency**	Enables instant profile fetch for web personalisation
Technical SLA	Abandoned Cart Alert SLA	**< 60 s post-threshold**	Triggers automated recovery email/SMS
Model SLA	Propensity Classifier AUC	**AUC >= 0.75 (F1 >= 0.65)**	Accurately targets high-intent buyers

3. END-TO-END SYSTEM ARCHITECTURE DIAGRAM

End-to-End System Architecture — Project 01: E-Commerce Customer 360

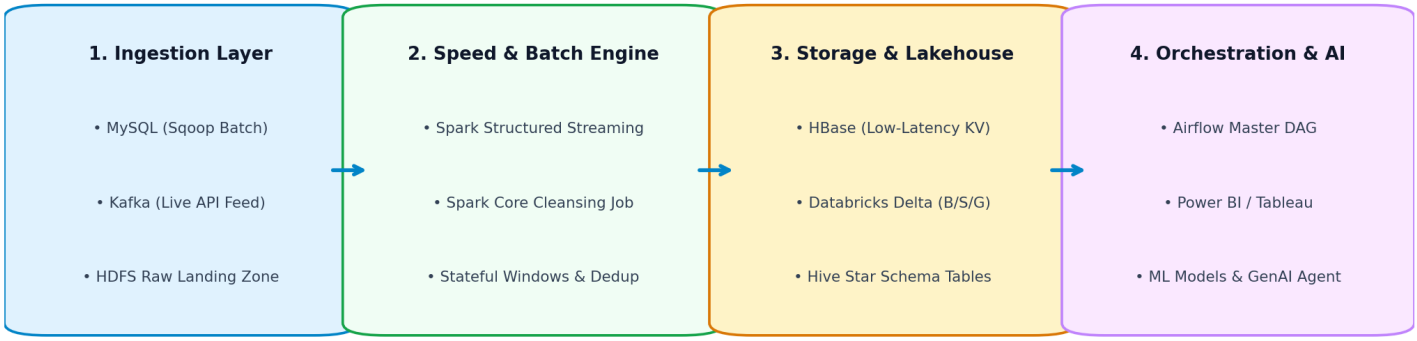


Figure 01.1: Four-layer technical architecture mapping batch/streaming ingestion, lakehouse storage, orchestration, and AI serving.

4. COMPREHENSIVE TECHNICAL MODULE SPECIFICATIONS

• **Ingestion & Streaming:** Sqoop incremental import (lastmodified) of orders, customers, and products from MySQL into HDFS. Live clickstream JSON pushed to Kafka topic `cart_events`. Public Bluesky Jetstream feed filtered by brand keywords and Open-Meteo weather feed ingested via Python producer.

• **Processing & Cleansing:** Spark Core parses raw web logs using a malformed-record accumulator; broadcast join attaches product details. Spark Structured Streaming computes 10-minute session windows sliding every 1 minute to flag idle carts (>15 mins).

• **Storage & Data Modeling:** Hive star schema (`fact_orders`, `dim_customer`, `dim_product`, `dim_date`) partitioned by `order_date`. HBase table keyed by `customer_id` storing latest LTV, last-seen, and active cart status.

• **Cloud & Lakehouse Stack:** AWS S3 raw zone, Glue Data Catalog, Athena ad-hoc queries. Databricks Medallion: Bronze (Auto Loader), Silver (SCD Type 2 for address/tier changes), Gold (`customer_360`, `abandoned_carts`).

• **Orchestration & Workflow:** Airflow master DAG running nightly Sqoop incremental import, Spark batch aggregation, Hive load, HBase sync, and notification trigger on success.

• **Analytics & AI Integration:** Power BI 3-page dashboard (Sales Funnel, Cohort Retention, Abandoned Carts). K-Means customer segmentation and Random Forest purchase propensity model. GenAI RAG chatbot over support tickets and reviews using vector store embeddings.

5. LIVE STREAM & REPLAY HYBRID DESIGN

Hybrid Kafka Producer Strategy

Live & Replay Architecture: Bluesky Jetstream WebSocket (brand mentions & sentiment) + Open-Meteo weather feed + Olist E-commerce / Retailrocket clickstream replay at 100x speed. This hybrid architecture ensures continuous 24/7 event generation while supporting historical backfills.

6. TARGET MEASURABLE OUTCOMES & ACCEPTANCE RUBRIC

Key Verification Outcomes: Cart alert triggered within 60s of idle threshold. HBase fetch <100ms. Live brand sentiment updated every 1 min on BI dashboard.

Common Failure Drills Executed: 1) Broker Kill Drill: Halting 1 of 3 Kafka brokers causes zero data loss. 2) Spark Restart Drill: Streaming job resumes from checkpoint with exact-once semantics. 3) Reconciliation Drill: Record counts match between Kafka, Bronze, Silver, and Gold Delta layers.